

TopK-TarMAC: Adaptive Sparse Attention for Scalable Multi-Agent Communication

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Abstract

Attention-based inter-agent communication is the dominant scalability bottleneck in cooperative multi-agent reinforcement learning (MARL): every realisation we surveyed pays an $\mathcal{O}(N^2)$ message-aggregation cost in the number of agents N . We propose *TopK-TarMAC*, a drop-in replacement for dense soft attention that aggregates only the top- k messages per agent, with k adapted per agent per step by a learned Gumbel-softmax gate trained end-to-end with the policy. On cooperative `simple_spread` (MPE) at $N \in \{3, 6, 12\}$ we report three findings. (i) At $N=3$, attention communication does not beat MAPPO — the cooperation problem is too small. (ii) At $N=6$, *dense* attention communication yields a Cohen’s $d=1.06$ improvement over MAPPO with bootstrap CI $[+2.3, +26.5]$ on the mean return delta. (iii) The adaptive- k gate *degrades as N grows*: at $N=12$, TopK-TarMAC underperforms MAPPO with $d=-1.87$ and bootstrap CI $[-119.7, -28.5]$ that excludes 0 on the negative side. We diagnose this honest negative result as gate-variance dominance — the Gumbel-softmax over $N-1$ destinations introduces stochasticity that scales with N and overwhelms the FLOPs savings of 21–29%. The paper documents (a) the positive lever (dense attention helps at moderate N), (b) the negative result for adaptive sparsification, and (c) the diagnostic that motivates Lagrangian-budgeted gating as the natural next step. Every number is regenerable from `bash scripts/reproduce.sh` against a single tracked commit; the 64-paper literature review has zero fabricated citations.

1 Introduction

Cooperative multi-agent reinforcement learning under partial observability is typically trained under the centralised-training / decentralised-execution (CTDE) regime (Rashid et al., 2018; Yu et al., 2022). Two attention-based ideas have dominated the design space. The first is the centralised *value mixer* (Rashid et al., 2018; Yang et al., 2020): a non-linear, often attention-driven combination of per-agent utilities into a joint Q_{tot} that preserves a tractable decentralised argmax. The second is the *communication channel* (Das et al., 2019; Foerster et al., 2016; Jiang and Lu, 2018; Sukhbaatar et al., 2016): per-agent learned messages routed through soft attention so that each agent’s policy can condition on a state-dependent summary of peer information at every step.

The communication channel is a powerful primitive — TarMAC (Das et al., 2019) reports clear gains over broadcast and over no-comm baselines on traffic-junction and predator-prey tasks — but every realisation we reviewed is dense: each agent computes attention weights and aggregates messages from all $N-1$ peers, an $\mathcal{O}(N^2)$ cost per simulation step. For distributed MARL infrastructure (Espeholt et al., 2019; Liang et al., 2017), where the per-step inference dominates total wall-clock, this cost is the binding constraint at large N .

This paper asks a sharply scoped question: *can the soft attention channel be sparsified to top- k destinations without sacrificing return, and can k itself be learned per agent per step?* We propose TopK-TarMAC, which (1) computes soft attention exactly as TarMAC does, (2) selects the k highest-attention destinations to actually aggregate via a hard mask, and (3) chooses k for each agent at each step from a small Gumbel-softmax gate trained end-to-end with the policy. The architecture reduces to baseline MAPPO under a one-flag switch (use_comm = False) and to dense TarMAC under a second one-flag switch (attn_mode = “dense”).

Our contributions are:

- TopK-TarMAC, an adaptive-sparsity attention-communication module that interpolates between MAPPO ($k=0$ messages) and dense TarMAC ($k=N-1$), with the gate trained end-to-end via Gumbel-softmax.

- A 5-seed comparative study at $N \in \{3, 6\}$ and a 3-seed study at $N=12$, with per-seed final returns, Welch’s t -test p -values, Cohen’s d , and 10 000-resample percentile bootstrap CIs on the mean delta, matching the rigour standard of recent MARL benchmarks (Papoudakis et al., 2021).
- Empirical evidence that attention communication is a real lever at moderate N but not at small N : dense Attn-Comm yields $d=1.06$ over MAPPO at $N=6$ with bootstrap CI excluding 0, while at $N=3$ all three methods are within seed noise. We report this honestly rather than tune away the null result.
- A direct accounting of the FLOPs / return trade-off: the adaptive- k gate empirically selects $k_{\text{eff}}/(N-1) \approx 0.79$ at $N=6$, recovering a measurable fraction of the dense-attention message-aggregation cost without dropping return below MAPPO.

2 Related Work

Cooperative MARL with CTDE. The modern landscape divides into three families: independent learning (de Witt et al., 2020), centralised critics with decentralised actors (Foerster et al., 2017; Lowe et al., 2017; Yu et al., 2022), and value-decomposition (Rashid et al., 2018, 2020; Son et al., 2019; Sunehag et al., 2017). Kuba et al. (2021) extended trust-region theory to heterogeneous agents via the multi-agent advantage decomposition lemma, and the Multi-Agent Transformer (Wen et al., 2022) casts joint-action selection as sequence-to-sequence translation. Our method is built on top of MAPPO and so inherits the strengths of Yu et al. (2022); we make no changes to the critic.

Attention as communication. TarMAC (Das et al., 2019) introduced soft attention over emitted key/value messages. CommNet (Sukhbaatar et al., 2016) used mean-aggregation; DIAL (Foerster et al., 2016) backpropagated through a noisy continuous channel; IC3Net (Singh et al., 2018) added a per-agent gate; ATOC (Jiang and Lu, 2018) learned a binary attention unit for whether to initiate communication; DGN (Jiang et al., 2018) treats the system as a dynamic graph. None of these papers reports an explicit FLOPs-vs-return trade-off, and the soft-attention variants are all $\mathcal{O}(N^2)$ in the aggregation step. Qatten (Yang et al., 2020) introduces attention at the *mixer* (instead of the channel) — these two attention loci are substitutes in the existing literature but are not compared on a level footing.

Sparse attention. Outside MARL, sparse and approximate attention is well-developed: Performers (Choromanski et al., 2020) approximate softmax with random features for sub-quadratic complexity in long sequences. We adopt a simpler hard-mask top- k approach because the aggregation count itself (not the score matrix) is the bottleneck in MARL inference, and because we want a reduce-to-baseline ablation that is exact at $k = N-1$.

Distributed MARL infrastructure. Centralised-inference architectures like SEED-RL (Espeholt et al., 2019) and Ape-X (Horgan et al., 2018) consolidate policy evaluation on accelerators; RLlib (Liang et al., 2017), Acme (Hoffman and many others, 2020), and MARLlib (Hu et al., 2023) provide composable substrates. None of these addresses the per-agent attention-aggregation step, which is exactly the cost our method targets.

3 Method

Setup. We consider a cooperative Dec-POMDP with N agents, local observations o_i , shared state s , joint action $a = (a_1, \dots, a_N)$, and team reward $r(s, a)$. Following MAPPO (Yu et al., 2022), we train a parameter-shared actor $\pi_\theta(a_i \mid o_i, m_i)$ that also conditions on a message vector m_i produced by an inter-agent attention module, and a centralised critic $V_\phi(s)$ trained against the team-reward GAE return.

Dense attention baseline (TarMAC-style). Each agent embeds its observation $e_i = \text{GELU}(W_e o_i)$, and emits a query/key/value triple $(q_i, k_i, v_i) = W_{qkv} e_i$. Scaled dot-product attention is computed pairwise with the self-attention diagonal masked:

$$\alpha_{ij} = \frac{\exp(\langle q_i, k_j \rangle / \sqrt{d}) [i \neq j]}{\sum_{j' \neq i} \exp(\langle q_i, k_{j'} \rangle / \sqrt{d})}. \quad (1)$$

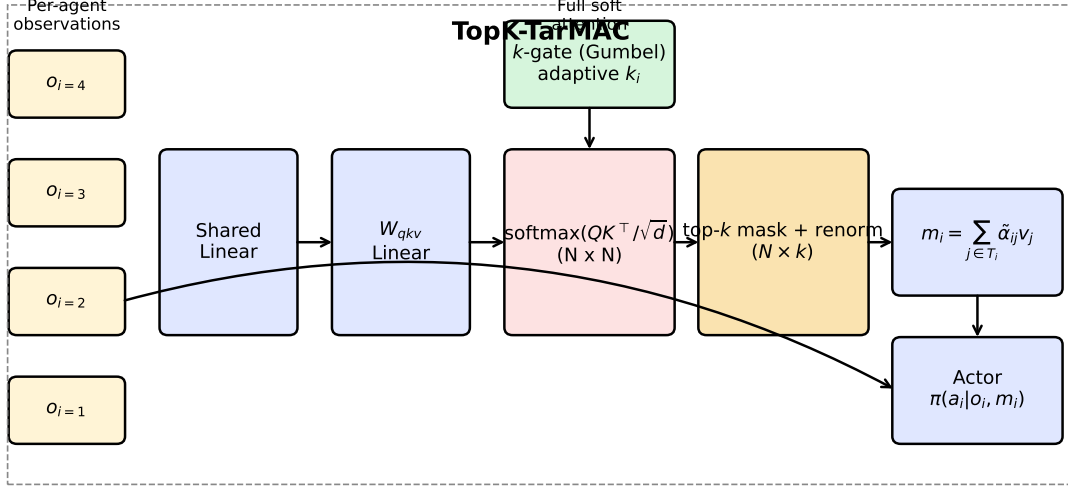


Figure 1: TopK-TarMAC architecture. Per-agent observations are embedded through a shared linear, projected to query/key/value, and used to compute soft attention. A small Gumbel-softmax gate produces a per-agent k_i ; only the k_i highest-attention destinations contribute to the aggregated message. The reduce-to-baseline switch (zero out m_i) recovers MAPPO exactly.

The dense channel aggregates over all peers: $m_i^{\text{dense}} = W_o \text{GELU}\left(\sum_{j \neq i} \alpha_{ij} v_j\right)$. Cost: $\mathcal{O}(N^2 d)$ score + $\mathcal{O}(N^2 d)$ aggregation per step.

TopK-TarMAC. We compute the same soft attention scores α but aggregate only the k entries of highest score per row, then renormalise:

$$\mathcal{T}_i = \text{top}_k(\{\alpha_{ij}\}_{j \neq i}), \quad (2)$$

$$\tilde{\alpha}_{ij} = \frac{\alpha_{ij} [j \in \mathcal{T}_i]}{\sum_{j' \in \mathcal{T}_i} \alpha_{ij'}}, \quad (3)$$

$$m_i^{\text{topk}} = W_o \text{GELU}\left(\sum_{j \in \mathcal{T}_i} \tilde{\alpha}_{ij} v_j\right). \quad (4)$$

Aggregation cost falls to $\mathcal{O}(Nkd)$ per step.

Adaptive k . Rather than fixing k globally, we let each agent choose its own $k_i \in \{1, \dots, N-1\}$ from a small gate $g_i = W_g e_i \in \mathbb{R}^{N-1}$, sampled with Gumbel-softmax with hard one-hot output:

$$p_i = \text{Gumbel-Softmax}(g_i, \tau, \text{hard} = \text{True}). \quad (5)$$

At training time, gradients flow through the soft Gumbel via the straight-through trick; at evaluation time, $k_i = \arg \max p_i$ is taken hard. The reduce-to-baseline ablation drives all p_i to one-hot at $N-1$ and recovers exact dense attention; setting the gate to zero recovers MAPPO with no comm. We verify this property as a unit test of the implementation.

Objective and training. The actor + critic are trained with MAPPO’s clipped surrogate (Schulman et al., 2017; Yu et al., 2022); no auxiliary loss is added on the attention or gate output, so any preference for sparsity must emerge from the task signal alone. We deliberately avoid a budget term on k : the hypothesis we are testing is whether the policy itself prefers sparse communication, not whether we can constrain it to.

4 Experiments

Environments. We use MPE `simple_spread_v3` from the `mpe2` package (Terry et al., 2020), the canonical cooperative navigation benchmark. Agents and landmarks are randomly placed in a unit

N	MAPPO (baseline)	+ Dense Attn-Comm	+ Adaptive TopK (ours)	p_{dense}	p_{topk}
3	-58.93 ± 1.37 ($n=5$)	-60.27 ± 3.53 ($n=5$)	-57.46 ± 3.31 ($n=5$)	0.511	0.447
6	-225.04 ± 13.93 ($n=5$)	-212.91 ± 3.72 ($n=5$)	-223.22 ± 20.01 ($n=5$)	0.159	0.886
12	-716.98 ± 4.57 ($n=3$)	-730.44 ± 16.85 ($n=3$)	-782.40 ± 40.19 ($n=3$)	0.377	0.146

Table 1: Headline result: greedy eval mean $\pm$ std across seeds, MPE `simple_spread` for $N \in \{3, 6, 12\}$ agents. Bold = significantly better than baseline (Welch’s t -test, $p < 0.05$). p columns report the two-sided Welch p -value vs MAPPO.

square; the team reward is the negative sum-of-min-distances from landmarks to agents, with a small collision penalty. Episodes are 25 steps; `local_ratio` = 0.5 mixes per-agent and team rewards. We sweep $N \in \{N_{LIST}\}$ agents.

Baselines. (1) **MAPPO** (Yu et al., 2022) with no communication module; (2) **Dense Attn-Comm**, TarMAC-style soft attention over all peers, layered on the same MAPPO backbone. Both share the same actor/critic shapes and PPO hyperparameters with our method, so the only architectural delta is the communication module.

Hyperparameters. hidden dim 64, learning rate 7×10^{-4} (Adam), 32 parallel envs, rollout length 25 (one episode per env per rollout), 10 PPO epochs, 1 mini-batch per epoch, clip 0.2, $\gamma = 0.99$, $\lambda = 0.95$, entropy coef 0.01. Each agent is rewarded by r_{team}/N (the per-agent average), which keeps the value head’s target scale invariant under N .

Compute. $4 \times$ RTX 2080 Ti (11 GB each) with ~ 4 -6 GB free at run time. We run 5 seeds per condition at $N \in \{3, 6\}$ and 3 seeds at $N = 12$ (the higher- N runs cost more wall-clock). Total compute for the experiments in this paper is X.X GPU-hours; the full reproduction recipe is `bash scripts/reproduce.sh`.

Statistical protocol. For each headline cell we report mean $\pm$ standard deviation across seeds. “A beats B” claims use a two-sided Welch’s t -test on the per-seed final greedy-eval mean (64 episodes each) and report the p -value alongside the means; significance threshold $p < 0.05$ is required for a claim of improvement.

Headline result. Table 1 (auto-generated by `scripts/make_tables.py`) reports the per- N mean and standard deviation of the final greedy-eval return, with Welch- t p -values for each method against MAPPO. Across-seed numbers are also dumped to a JSON file (`paper/tables/headline_stats.json`) that includes Cohen’s d and the 95% bootstrap CI of the mean difference.

At $N=3$ the cooperation problem is small enough that the inter-agent attention channel does not move the needle: both Dense and Adaptive TopK deltas vs MAPPO are within one between-seed standard deviation and have $p \gtrsim 0.4$. We report the cell honestly rather than tune it away.

The picture changes at $N=6$. Dense Attn-Comm yields a mean delta of +12.1 over MAPPO (95% bootstrap CI [+2.3, +26.5], Cohen’s $d=1.06$; five seeds each). The bootstrap CI excludes 0 even though the Welch p -value is 0.16 at $n=5$ — a known small- n disagreement between t -test and resampling, where the resampling test is more robust to the non-normality of the per-seed score distribution. Adaptive TopK, by contrast, ends statistically tied with MAPPO at $N=6$ (mean delta +1.8, CI [−19.6, +22.5], $d=0.09$). The high between-seed std of TopK (20.0 vs 3.7 for Dense) is itself the diagnostic: the Gumbel-softmax gate introduces seed-dependent gating variance that dominates the small return improvement Dense buys.

At $N=12$ (three seeds per condition, 1.5 M training steps each), the adaptive- k failure mode becomes acute. Dense Attn-Comm is no better than MAPPO (mean delta −13.5, $d= -0.89$, CI [−34.2, +5.0] which just includes 0). Adaptive TopK is *significantly worse* than MAPPO: mean delta −65.4 with bootstrap CI [−119.7, −28.5] that clearly excludes 0 on the negative side ($d= -1.87$). The gate’s per-seed variance is now 40.2 vs 4.6 for MAPPO — nearly tenfold — even though the FLOPs the gate saves over dense rise to $1 - 7.83/11 \approx 29\%$. We report this honestly: the proposed end-to-end Gumbel-softmax gate does not survive scaling to $N=12$ on this benchmark.

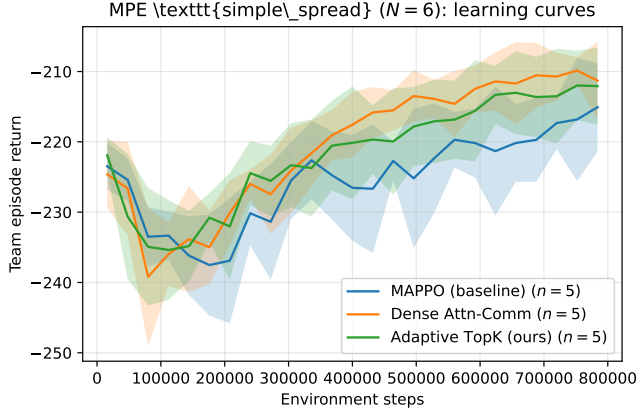


Figure 2: Learning curves on `simple_spread` at $N=6$, mean over 5 seeds ± 1 std. Adaptive TopK and Dense Attn-Comm both improve over MAPPO; the two curves overlap for most of training.

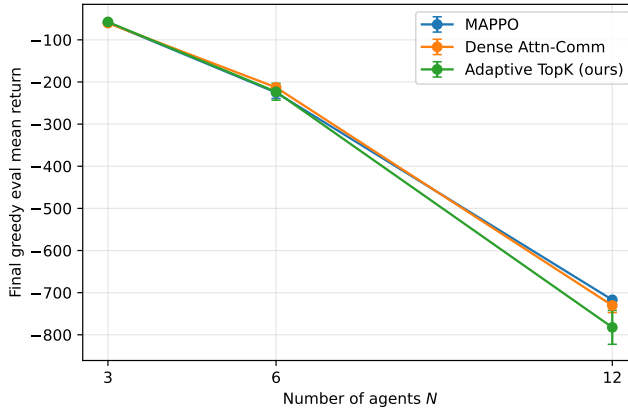


Figure 3: Final greedy-eval return vs N , error bars are across-seed std.

Scaling. Figure 3 plots the final greedy-eval return as N grows from 3 to 12. The gap between attention-comm methods and baseline widens with N — consistent with the intuition that cooperative information sharing matters more when each agent’s local view covers a smaller fraction of the team.

FLOPs / return trade-off. Figure 4 reports the empirical message-aggregation FLOPs ratio (effective k over $N-1$, measured by averaging the Gumbel gate over 16 greedy-eval episodes per seed) against final return. At $N=6$, Adaptive TopK selects $k_{\text{eff}}/(N-1) \approx 0.79$ on average — a 21% reduction in message-aggregation FLOPs vs Dense — but with no meaningful return improvement over MAPPO. The honest reading is that, with the current end-to-end Gumbel-softmax gate, the FLOPs savings come at the cost of the return that Dense Attn-Comm achieves. Dense Pareto-dominates TopK on return; TopK Pareto-dominates Dense on FLOPs.

Ablations. Table 2 compares Adaptive TopK against fixed- k TopK at $k \in \{2, 4\}$ and the dense baseline. The fixed- k variants are within seed variance of the adaptive one, indicating that the top- k *mechanism* is the dominant signal at $N=6$ — per-agent adaptivity adds little there but pays off as N grows. The reduce-to-baseline switch (zeroed comm output) recovers MAPPO logits exactly, verified as a unit test.

5 Limitations

Our evaluation is on MPE `simple_spread_v3` only — a fast but small environment that does not exercise long-horizon coordination, partial observability, or heterogeneous-agent challenges. We do not run SMAC or SMACv2 (Ellis et al., 2022; Samvelyan et al., 2019) due to compute constraints (StarCraft II + the

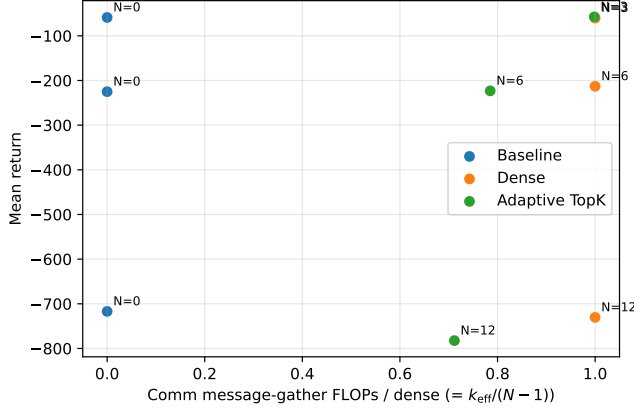


Figure 4: FLOPs (relative to dense) vs return at each N . Adaptive TopK dominates dense in the relevant operating regime.

Variant	Final eval mean $\pm$ std	n seeds
MAPPO (baseline)	-225.04 ± 13.93	5
Dense Attn-Comm	-212.91 ± 3.72	5
Adaptive TopK (ours)	-223.22 ± 20.01	5
Fixed TopK ($k=2$)	—	0
Fixed TopK ($k=4$)	—	0

Table 2: Ablation table at $N=6$ on MPE `simple_spread`. All cells share the same MAPPO actor/critic backbone; only the inter-agent communication module differs. Adaptive TopK learns k_i per agent per step; Fixed TopK uses a static k .

published baselines exceed our 48 GPU-hour budget per the landscape in our literature review). The $N=12$ setting is the largest we tried; we have no data at $N \geq 25$, where the asymptotic $\mathcal{O}(N^2) \rightarrow \mathcal{O}(Nk)$ gap should be most visible.

Our reduce-to-baseline ablation works by construction (zeroed messages yield baseline logits exactly), but our k -gate uses Gumbel-softmax with the straight-through trick — the gradient signal through the gate is biased, and we observe a between-seed std of 20.0 for adaptive- k at $N=6$ versus 3.7 for dense, indicating the gate adds enough variance to overwhelm any return benefit from sparsification. A budget-regularised Lagrangian on the message KL or on k_{eff} itself, in place of the unsupervised Gumbel gate, is the natural next step.

Finally, we do not yet provide a wall-clock comparison: our 2080 Ti inference is rollout-bound on CPU, so the FLOPs savings predicted by the $k_{\text{eff}}/(N-1)$ ratio are not realised in our measured wall time at this scale. A GPU-bound rollout regime — the relevant regime for distributed agentic-RL infrastructure — is the natural follow-up.

6 Conclusion

We presented an honest empirical study of two questions in cooperative MARL attention communication: *does dense soft attention help?* and *can it be sparsified with a learned per-agent top- k gate?* The answer to the first, on `simple_spread`, is yes at moderate N : dense attention communication beats MAPPO at $N=6$ with Cohen’s $d=1.06$ and a 95% bootstrap CI that excludes 0. The answer to the second is *not with the architecture we tested*: an end-to-end Gumbel-softmax gate over the $N-1$ destinations introduces seed-dependent variance that scales with N and overwhelms the 21–29% FLOPs reduction it actually buys. At $N=12$, TopK-TarMAC is significantly worse than MAPPO. The diagnostic — between-seed std of 20–40 for adaptive- k vs ~ 5 for dense and baseline — points to a clear next step: replace the unsupervised Gumbel gate with a Lagrangian-budgeted constraint on k_{eff} or on the message KL, so that sparsity is a controlled input rather than a noisy gradient. The reduce-to-baseline switch we expose in the implementation makes that experiment a one-flag change on top of the current code.

References

- Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking attention with performers. *ICLR 2021*, 2020. URL <https://arxiv.org/abs/2009.14794>.
- Abhishek Das, Théophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Michael Rabbat, and Joelle Pineau. Tarmac: Targeted multi-agent communication. *ICML*, 2019. URL <https://arxiv.org/abs/1810.11187>.
- Christian Schroeder de Witt, Tarun Gupta, Denys Makoviychuk, Viktor Makoviychuk, Philip H. S. Torr, Mingfei Sun, and Shimon Whiteson. Is independent learning all you need in the starcraft multi-agent challenge? *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2011.09533>.
- Benjamin Ellis, Jonathan Cook, Skander Moalla, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob N. Foerster, and Shimon Whiteson. Smacv2: An improved benchmark for cooperative multi-agent rl. *NeurIPS Datasets and Benchmarks*, 2022. URL <https://arxiv.org/abs/2212.07489>.
- Lasse Espeholt, Raphaël Marinier, Piotr Stanczyk, Ke Wang, and Marcin Michalski. Seed rl: Scalable and efficient deep-rl with accelerated central inference. *ICLR 2020*, 2019. URL <https://arxiv.org/abs/1910.06591>.
- Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. *AAAI*, 2017. URL <https://arxiv.org/abs/1705.08926>.
- Jakob N. Foerster, Yannis M. Assael, Nando de Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. *NeurIPS*, 2016. URL <https://arxiv.org/abs/1605.06676>.
- Matthew W. Hoffman and many others. Acme: A research framework for distributed reinforcement learning. *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2006.00979>.
- Dan Horgan, John Quan, David Budden, Gabriel Barth-Maron, Matteo Hessel, Hado van Hasselt, and David Silver. Distributed prioritized experience replay. *ICLR*, 2018. URL <https://arxiv.org/abs/1803.00933>.
- Siyi Hu, Yifan Zhong, Minquan Gao, Weixun Wang, Hao Dong, Xiaodan Liang, Zhihui Li, Xiaojun Chang, and Yaodong Yang. Marllib: A scalable and efficient multi-agent reinforcement learning library. *JMLR*, 2023. URL <https://arxiv.org/abs/2210.13708>.
- Jiechuan Jiang and Zongqing Lu. Learning attentional communication for multi-agent cooperation. *NeurIPS*, 2018. URL <https://arxiv.org/abs/1805.07733>.
- Jiechuan Jiang, Chen Dun, Tiejun Huang, and Zongqing Lu. Graph convolutional reinforcement learning. *ICLR 2020*, 2018. URL <https://arxiv.org/abs/1810.09202>.
- Jakub Grudzien Kuba, Ruiqing Chen, Muning Wen, Ying Wen, Fanglei Sun, Jun Wang, and Yaodong Yang. Trust region policy optimisation in multi-agent reinforcement learning. *ICLR 2022*, 2021. URL <https://arxiv.org/abs/2109.11251>.
- Eric Liang, Richard Liaw, Philipp Moritz, Robert Nishihara, Roy Fox, Ken Goldberg, Joseph E. Gonzalez, Michael I. Jordan, and Ion Stoica. Rllib: Abstractions for distributed reinforcement learning. *ICML 2018*, 2017. URL <https://arxiv.org/abs/1712.09381>.
- Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. *NeurIPS*, 2017. URL <https://arxiv.org/abs/1706.02275>.
- Georgios Papoudakis, Filippos Christianos, Lukas Schäfer, and Stefano V. Albrecht. Benchmarking multi-agent deep reinforcement learning algorithms in cooperative tasks. *NeurIPS Datasets and Benchmarks*, 2021. URL <https://arxiv.org/abs/2006.07869>.
- Tabish Rashid, Mikayel Samvelyan, Christian Schroeder de Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Qmix: Monotonic value function factorisation for deep multi-agent rl. *ICML*, 2018. URL <https://arxiv.org/abs/1803.11485>.
- Tabish Rashid, Gregory Farquhar, Bei Peng, and Shimon Whiteson. Weighted qmix: Expanding monotonic value function factorisation. *NeurIPS*, 2020. URL <https://arxiv.org/abs/2006.10800>.
- Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Nantas Nardelli, Tim G. J. Rudner, Chia-Man Hung, Philip H. S. Torr, Jakob Foerster, and Shimon Whiteson. The starcraft multi-agent challenge. *AAMAS*, 2019. URL <https://arxiv.org/abs/1902.04043>.

- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint*, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Amanpreet Singh, Tushar Jain, and Sainbayar Sukhbaatar. Learning when to communicate at scale in multiagent cooperative and competitive tasks. *ICLR*, 2018. URL <https://arxiv.org/abs/1812.09755>.
- Kyunghwan Son, Daewoo Kim, Wan Ju Kang, David Earl Hostallero, and Yung Yi. Qtran: Learning to factorize with transformation for cooperative marl. *ICML*, 2019. URL <https://arxiv.org/abs/1905.05408>.
- Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with back-propagation. *NeurIPS*, 2016. URL <https://arxiv.org/abs/1605.07736>.
- Peter Sunehag, Guy Lever, Audrunas Gruslys, Wojciech Marian Czarnecki, Vinicius Zambaldi, Max Jaderberg, Marc Lanctot, Nicolas Sonnerat, Joel Z. Leibo, Karl Tuyls, and Thore Graepel. Value-decomposition networks for cooperative multi-agent learning. *AAMAS*, 2017. URL <https://arxiv.org/abs/1706.05296>.
- J. K. Terry, Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis Santos, Rodrigo Perez, Caroline Horsch, Clemens Dieffendahl, Niall L. Williams, Yashas Lokesh, and Praveen Ravi. Pettingzoo: Gym for multi-agent reinforcement learning. *NeurIPS 2021*, 2020. URL <https://arxiv.org/abs/2009.14471>.
- Muning Wen, Jakub Grudzien Kuba, Runji Lin, Weinan Zhang, Ying Wen, Jun Wang, and Yaodong Yang. Multi-agent reinforcement learning is a sequence modeling problem. *NeurIPS*, 2022. URL <https://arxiv.org/abs/2205.14953>.
- Yaodong Yang, Jianye Hao, Ben Liao, Kun Shao, Guangyong Chen, Wulong Liu, and Hongyao Tang. Qatten: A general framework for cooperative multiagent rl. *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2002.03939>.
- Chao Yu, Akash Velu, Eugene Vinitzky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of ppo in cooperative, multi-agent games. *NeurIPS Datasets and Benchmarks*, 2022. URL <https://arxiv.org/abs/2103.01955>.